ARCHITECTURE DEVELOPMENT PART 2: CONFIGURATION, INSATALLATION AND SECURITY OF OPENEMR.

A. Web page screenshot *(the screenshort must include the IP_Address of each hospital)* of the each hospital's successful installation of OpenEMR (8 points)

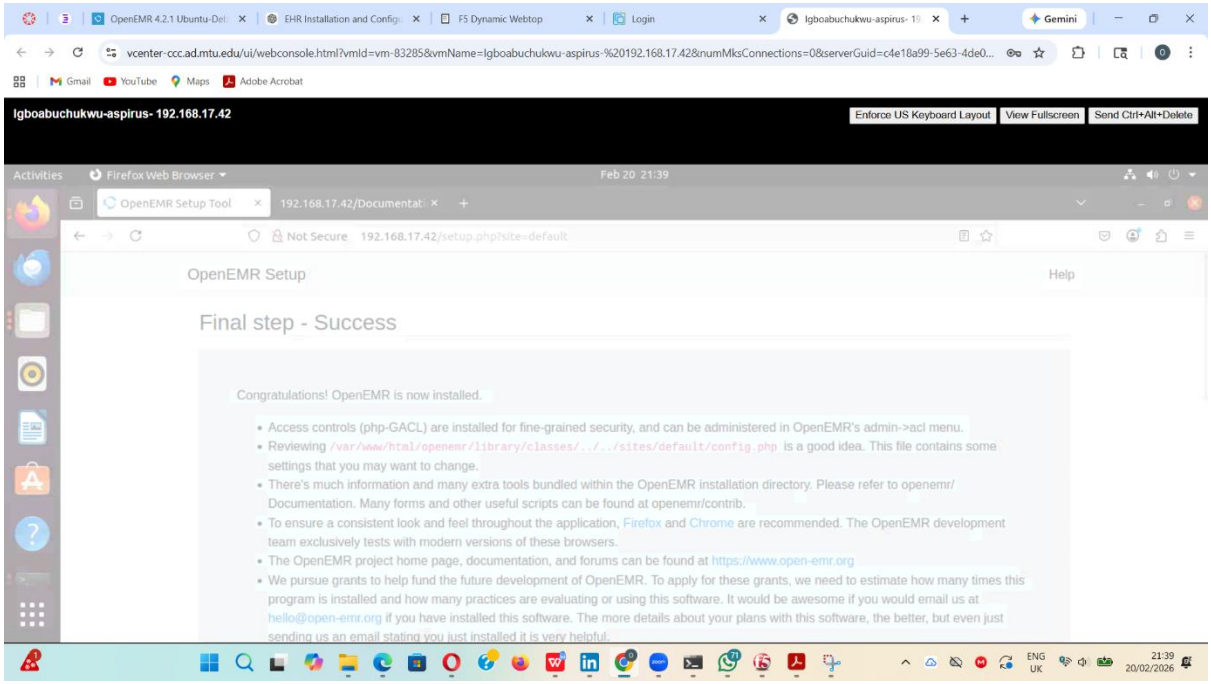


Fig 1: OpenEmr Success IP address: 192.168.17.42 – Aspirus Hospital

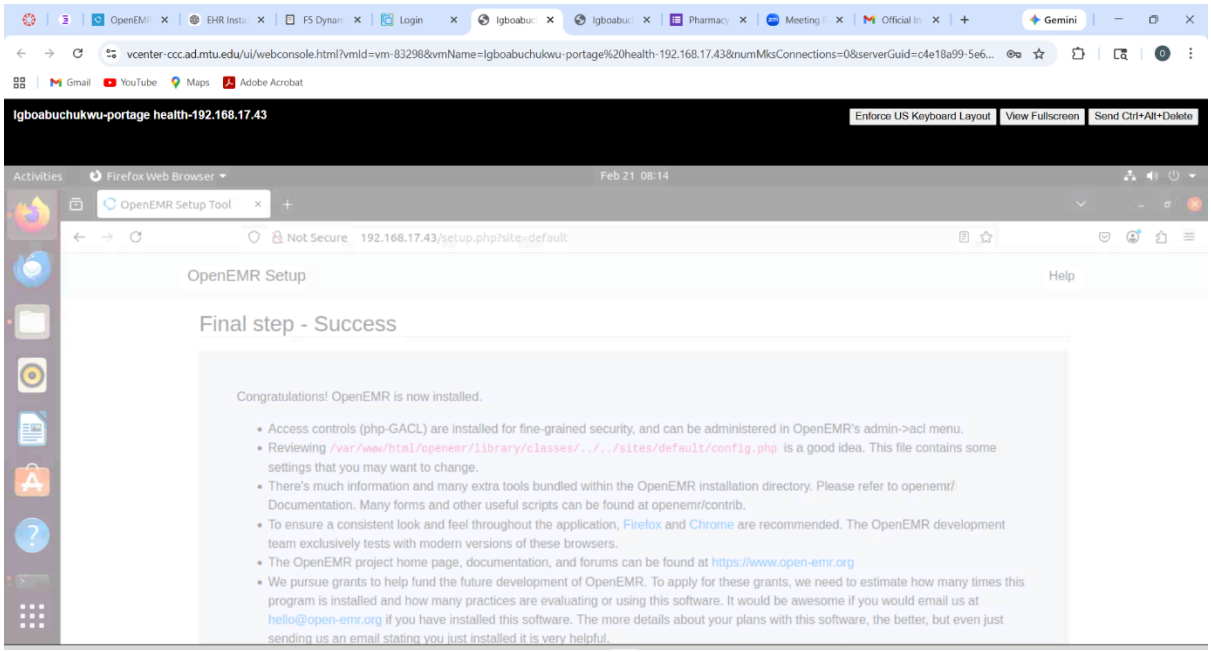


Fig 2: OpenEmr Success IP address: 192.168.17.43 – Portage Health Hospital

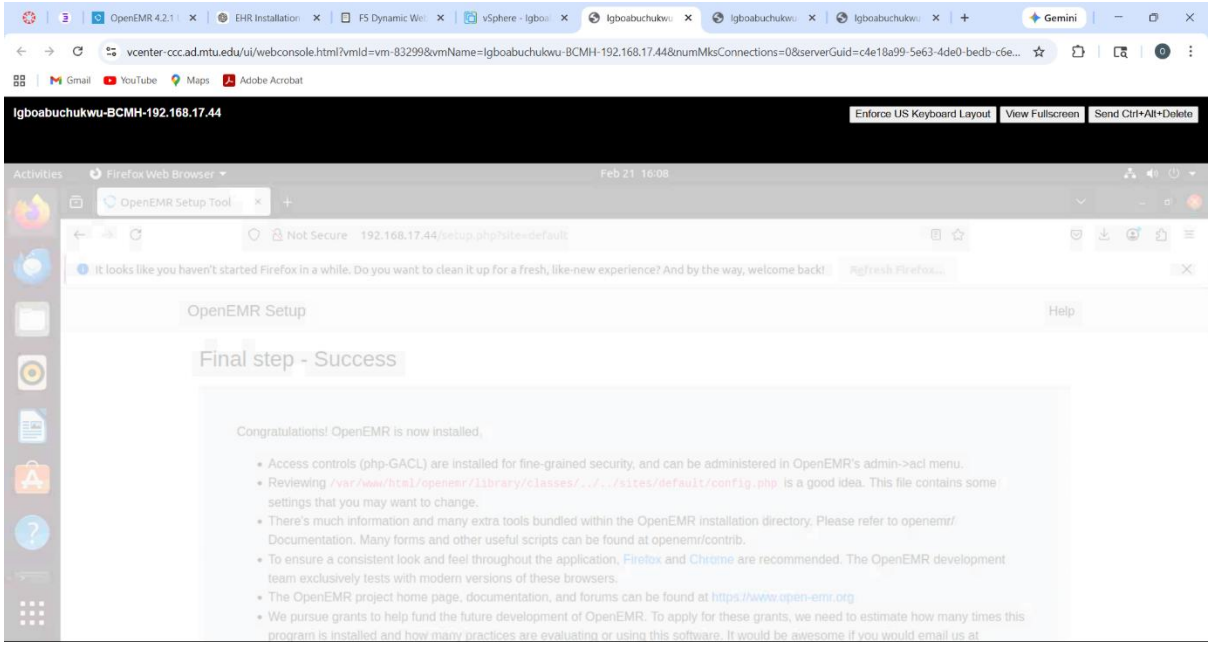


Fig 3: OpenEmr Success IP address: 192.168.17.44 – Baraga County Memorial Hospital

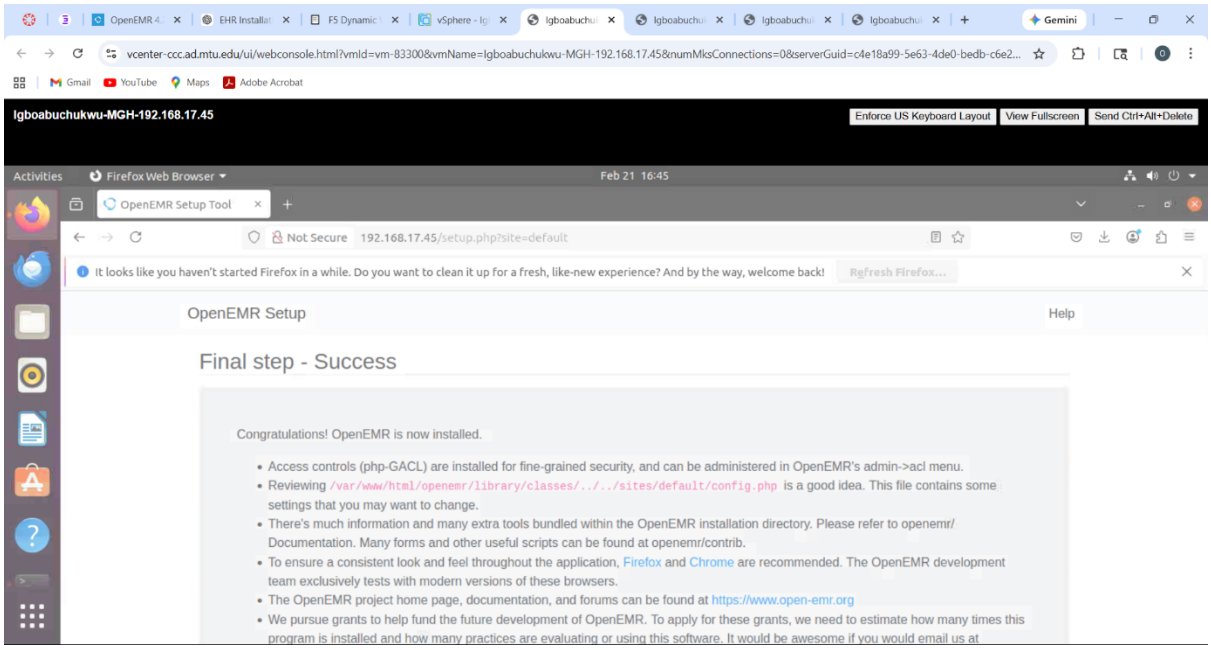
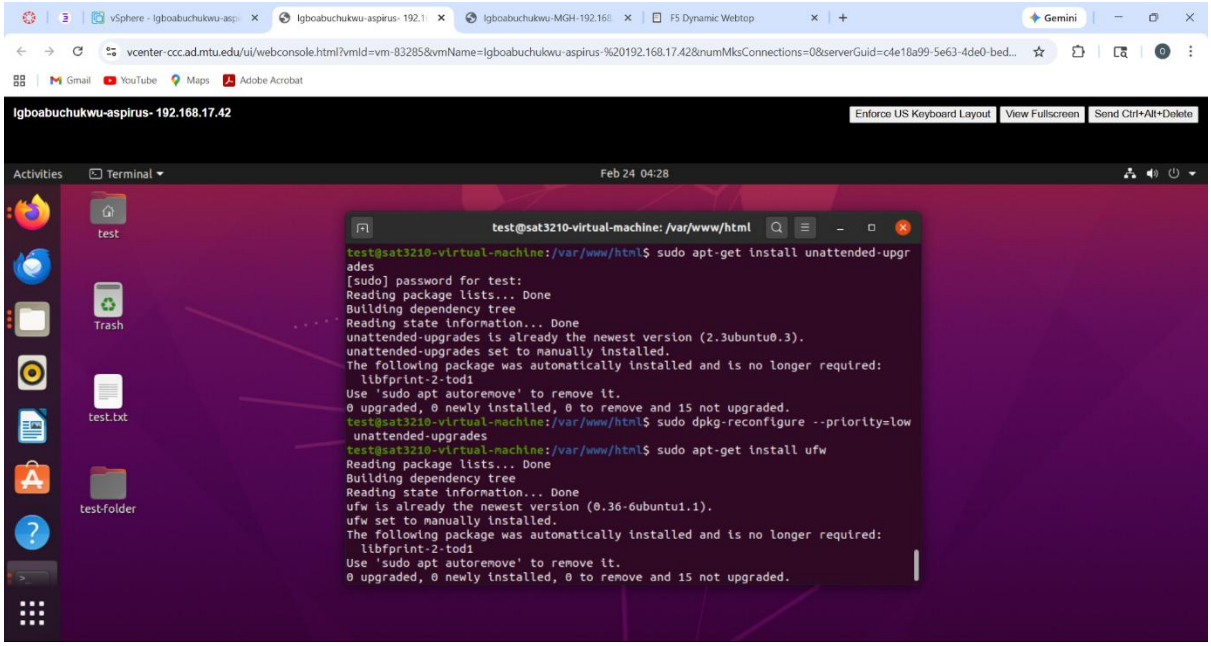


Fig 4: OpenEmr Success IP address: 192.168.17.45 –Marquette General Hospital

B. Show the steps and commands you used to secure OpenEMR (10 points)



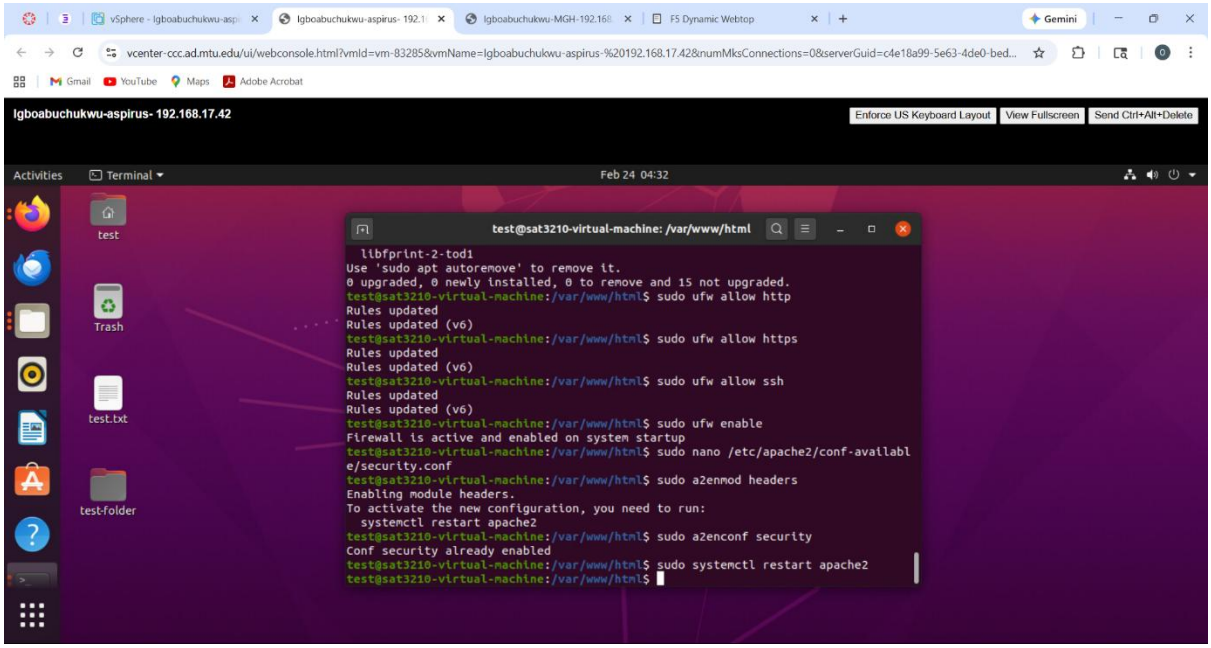
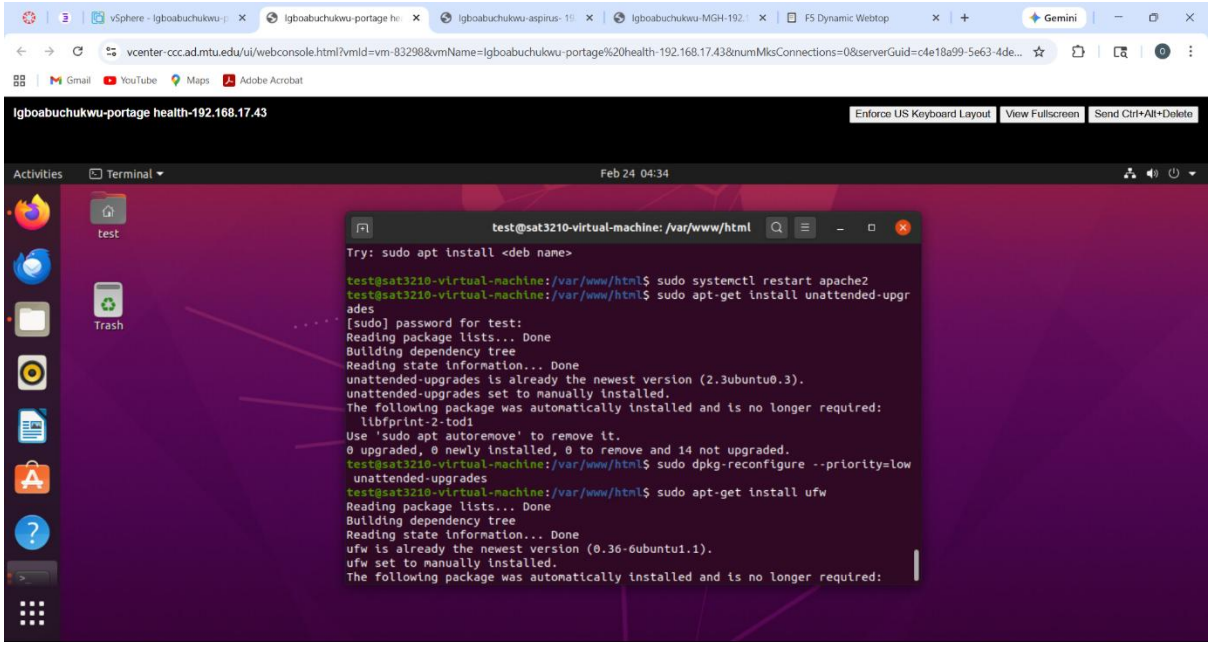


Fig 1: Steps showing the command for security configuration on Aspirus Hospital (192.168.17.42)



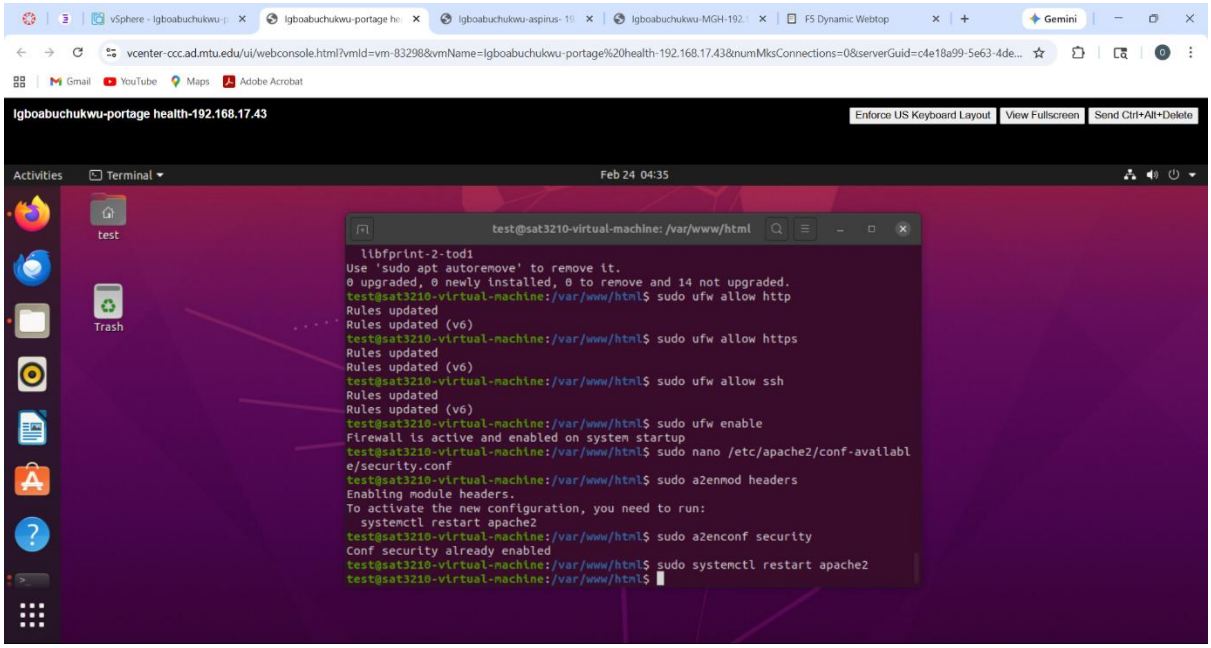
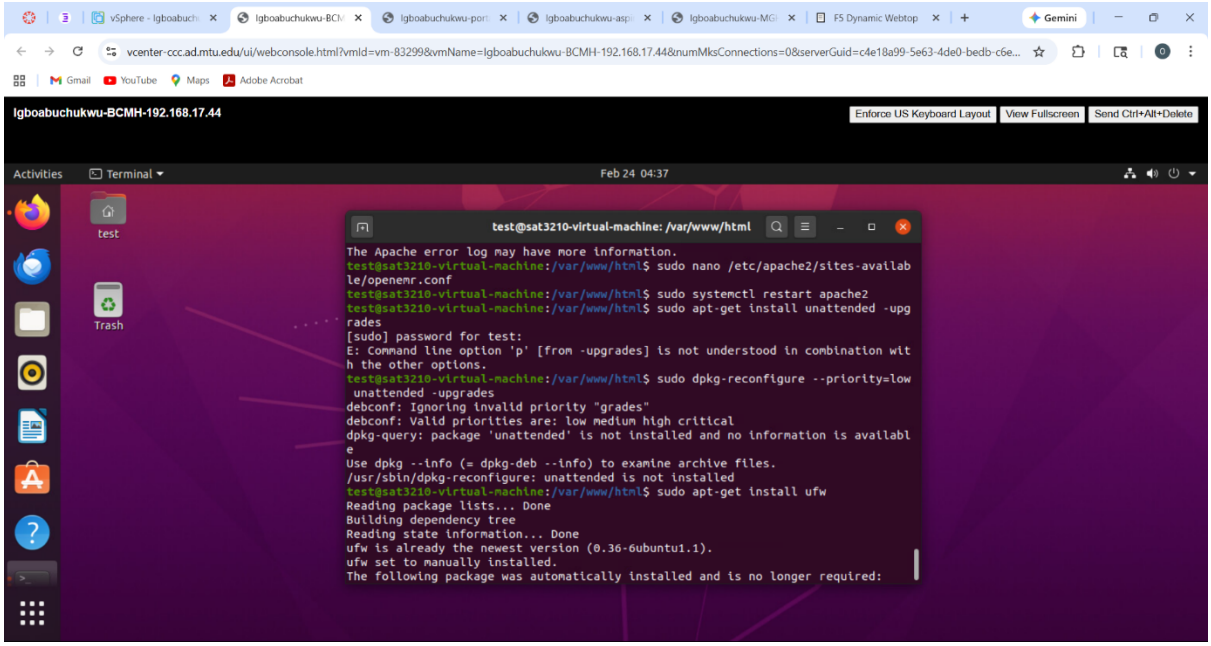


Fig 2: Steps showing the command for security configuration on Portage Health Hospital (192.168.17.43)



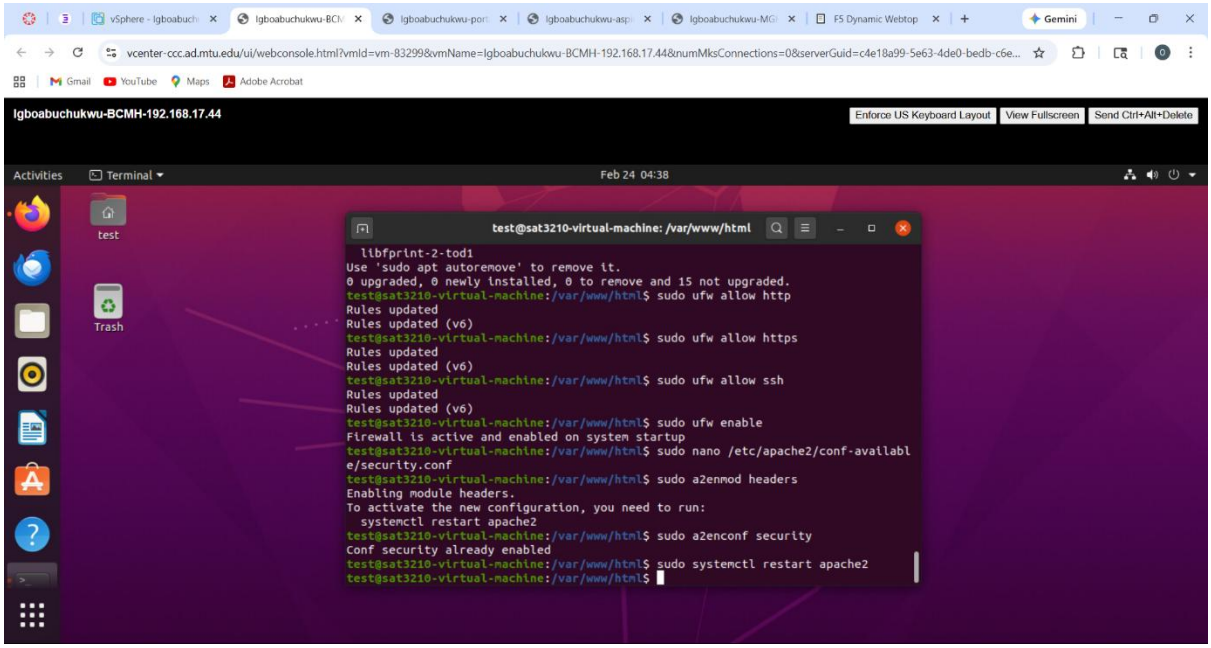
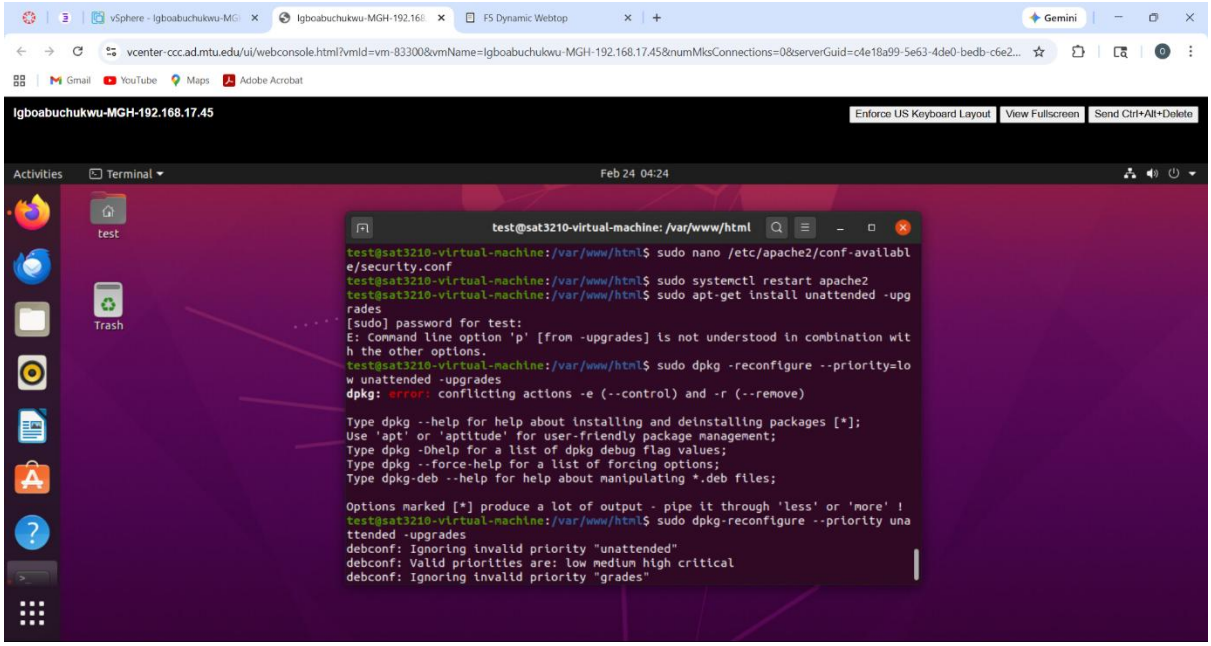


Fig 3: Steps showing the command for security configuration on Baraga County Memorial Hospital (192.168.17.44)



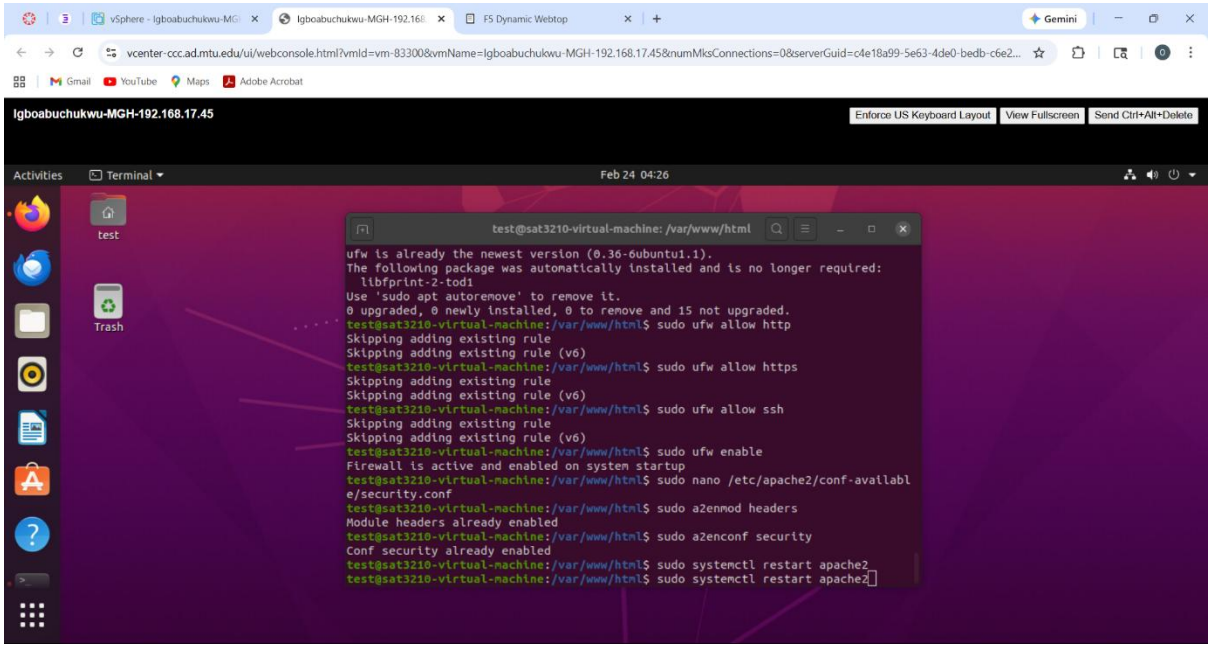


Fig 4: Steps showing the command for security configuration on Marquette General Hospital (192.168.17.45)

C. What other types of attacks would still be sucesptible to the OpenEMR platform? (2 points)

According to the OpenEMR Security Assessment documentation, several defensive protections are discussed, including risk management, access control, physical security, backup and disaster recovery, and network security. If these protections are not properly implemented or maintained, OpenEMR may become vulnerable to various types of attacks.

Weak Access Controls:
If authentication measures are inadequate, the system may be susceptible to **phishing attacks**,

where attackers trick users into revealing their login credentials, allowing unauthorized access to patient data.

Weak Physical Security:
If physical safeguards are lacking, the system may be exposed to **insider theft**, where an employee or authorized individual copies or removes sensitive patient information without permission.

Poor Backup and Disaster Recovery:
Without secure and properly tested backups, the system may be vulnerable to **ransomware attacks**, in which attackers encrypt live data and backup files, potentially disrupting patient care and causing data loss.

Weak Network Security:
If network protections such as firewalls and secure configurations are not enforced, the system may be susceptible to **SQL injection attacks**, where attackers manipulate database queries to access, modify, or delete sensitive information.